Measuring Library and Community Resilience

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Introduction

Problem: Rural communities face recurring environmental shocks that disrupt local institutions and population stability.

Gap: Existing research focuses on economic impacts but neglects institutional mediators.

Core Idea: Public libraries are uniquely positioned as multi-capital institutions, spanning social, human, financial, cultural, political, built, and natural systems.



The primary objectives of this study are:
- Do extreme events change library capacity?
- How do funding channels respond differently?
- Do libraries mediate migration outcomes?

Data Source & Preparation

Multiple Data Integration:

Data Sources	Variable Construction
Public Library Survey (PLS)	Library staffing, funding operating hours
FEMA Disaster Declarations	Timing and location of extreme events
IRS Statistics of Income (SOI)	County-level migration flows
Drought Data (PDSI)	Continous measure of environmental stress
Integrated Public Use Microdata Series (IPUMS)	Migration patterns

Table 1. *Variable and data source for library and county-level used in the event study model.*

Data Notes and Details:

(1) *A balanced panel dataset was constructed in Stata, consisting of 284,280 observations spanning the years 2003–2022.*

(2) *Observations from U.S. territories (e.g., Puerto Rico and the U.S. Virgin Islands) were excluded from the analysis.*

(3) *County definitions were harmonized for Connecticut and Alaska to ensure consistent geographic units over time.*

(4) *Counties are classified as ever treated if they experience at least one extreme event during the study period.*

(5) *An extreme event is defined as a FEMA-declared disaster, excluding drought declarations.*

Control Variables	Source
Population of Legal Service Area (Log)	PLS
Number of Branch Libraries (Log)	PLS
Number of Central Libraries (Log)	PLS
County Population	IPUMS
Poverty Rate	IPUMS
Unemployment Rate	IPUMS
Mean Income	IPUMS
Residential Mobility	IPUMS
County Migration	IRS-SOI

Table 2. *Variable and data source for library and county-level covariates used in the event study model.*

Methodology

Environmental shocks affect both institutional capacity and community outcomes. Public libraries are modeled as both affected entities and mediating mechanisms in this relationship.

Model 1: Institutional Impact

Model 1 measures impact of shocks on library capacity.

$$Y_{ijt} = \beta_1 xExtremeEvent_{it} + X_{ijt} + Z_{it} + \mu_i + \lambda_t + \epsilon_{ijt} \quad \text{(equation 1)}$$

Model 2: Event Study

$$Y_{ijt} = \alpha + \sum_{\tau=-1}^{\tau} \beta_\tau D_{it}^{\tau} + X_{ijt} + Z_{it} + \mu_i + \lambda_t + \varepsilon_{ijt} \quad \text{(equation 2)}$$

Model 2 compares library outcomes before and after disasters, controlling for county traits and national trends.

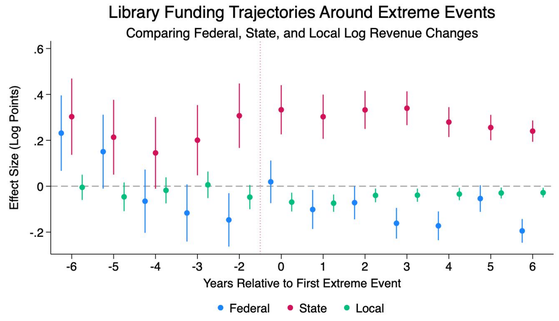
Model 3: Mediation

Model 3 tests whether libraries mediate community outcomes.

$$Y_{it} = \beta_0 + \beta_1 xExtremeEvent_{it} + \beta_2 xPublicLibrary_{it} X_{ijt} + Z_{it} + \epsilon_{ijt} \quad \text{(equation 3)}$$



Figure 1. *Library Funding*

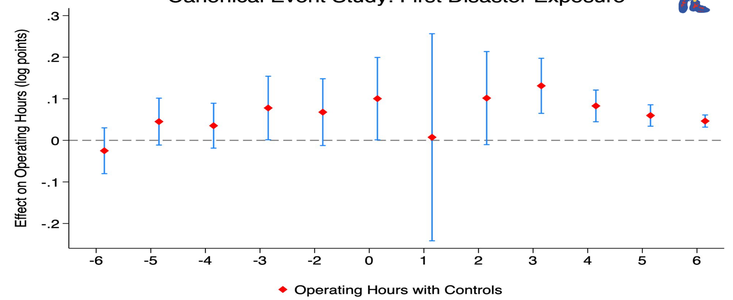


Insights 💡 *: State funding reacts immediately; however,* Federal Funding exhibits delayed adjustment. Pre-trends are unstable.

Results

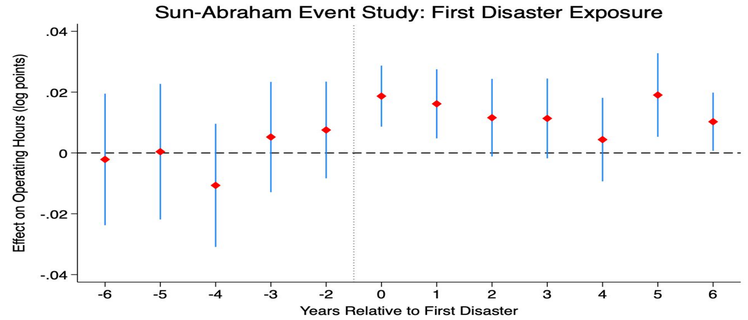
How Do Libraries Respond to Environmental Shocks?

Figure 2. *Outcome: Operating Hours (Canonical Model)*



Insights 💡 *:* Operating hours increase post-shock; estimates may be biased due improper weighting of counterfactual units.

Figure 3. *Outcome: Operating Hours (Sun-Abraham Model)*



Insights 💡 *:* Operating hours increase post-shock, effect ameliorates with time; contribution of each unit to estimate adjusted using Sun-Abraham.

Implications

More work can be done to examine the effect of extreme events on library capacity. Early evidence suggests changes in funding and operating hours. To truly measure community resilience, additional work should examine libraries as mediating factors for extreme events on community outcomes.